

Global Climate Indicators Dashboard Report

Research Question

This project investigates the relationship between atmospheric carbon dioxide (CO₂) emissions and global temperature anomalies over time. The primary goal is to determine whether increases in CO₂ emissions are associated with long-term changes in global temperature and how this relationship varies across different historical periods, including early industrialization, post–World War II expansion, and the modern era of accelerated climate change.

The analysis focuses on identifying long-term trends rather than short-term fluctuations, using interactive visualizations to better understand how human activity may be linked to changes in the Earth's climate system.

Data and Methodology

This dashboard combines two datasets: global CO₂ emissions per capita and Northern Hemisphere temperature anomalies (J-D index). These datasets were merged using the common variable of Year, creating a unified time series spanning from the late 19th century to the present.

To improve trend interpretation, both raw values and 5-year rolling averages were calculated. The rolling averages help reduce short-term variability and highlight long-term climate patterns more clearly.

The dashboard includes multiple interactive Plotly visualizations:

- Line charts showing CO₂ and temperature trends over time
- A scatter plot with a regression trendline showing the relationship between CO₂ and temperature
- A dual-axis chart comparing both variables on the same timeline
- A comparative analysis of pre- and post-1950 climate conditions
- Time range filtering across three historical periods

Key Findings

The analysis reveals several important climate trends.

First, global temperature anomalies show a consistent long-term upward trend. While there is some year-to-year variation, the overall pattern indicates increasing global temperatures, particularly after the mid-20th century. This suggests that warming is persistent rather than temporary.

Second, CO₂ emissions remain relatively stable in the early part of the dataset (1880–1949) but begin to rise sharply after 1950. This period corresponds with rapid industrial growth, increased fossil fuel consumption, and global economic expansion. The post-1950 era represents a major shift in human impact on the atmosphere.

Third, there is a strong positive relationship between CO₂ emissions and temperature anomalies. As CO₂ levels increase, temperature anomalies also tend to rise. The regression analysis in the dashboard supports

this relationship, showing a clear upward trend between the two variables. While this does not prove causation, it strongly suggests a meaningful connection between emissions and warming.

Fourth, the modern period (2000–present) shows the fastest rate of increase in both CO₂ and temperature. This indicates that climate change is not only ongoing but appears to be accelerating in recent decades.

Finally, a comparison between pre-1950 and post-1950 periods shows a clear structural difference. Both average CO₂ emissions and temperature anomalies are significantly higher after 1950, reinforcing the idea that industrial activity has had a lasting impact on the global climate system.

Limitations

Although the results show strong relationships between CO₂ emissions and temperature changes, several limitations should be considered.

First, the CO₂ dataset is aggregated, meaning it represents average values rather than country-specific or sector-specific emissions. This reduces detail and may mask important regional differences.

Second, the temperature data is based on anomalies rather than absolute temperature values. While anomalies are useful for detecting trends, they depend on a reference baseline and may not fully reflect localized climate variations.

Third, this analysis is observational and does not establish causation. While CO₂ and temperature are strongly correlated, other factors such as solar radiation, ocean cycles, and volcanic activity also influence global climate patterns.

Finally, the use of rolling averages improves clarity but can smooth out short-term fluctuations, potentially hiding sudden changes in the data.

Conclusion

Overall, the dashboard provides clear visual and statistical evidence of a strong relationship between CO₂ emissions and global temperature anomalies. Both variables show long-term increases, with particularly rapid growth occurring after 1950.

While the analysis has limitations in terms of data aggregation and causal inference, the consistent trends across multiple visualizations support the conclusion that rising CO₂ emissions are closely associated with global warming patterns.

This project demonstrates how interactive data visualization can be used to explore complex environmental relationships and communicate climate trends in an accessible and meaningful way.

Link to GitHub

<https://github.com/Da125673/Project>